

National Exam of Higher National Diploma 2026 Session

Specialty/option: SOFTWARE ENGINEERING (SWE), DATA BASE MANAGEMENT (DBM)
Paper: Information Systems
Duration: 3 hours

Credits: 5,6

GENERAL INSTRUCTIONS.

- **ANSWER ALL QUESTIONS**
- *You are reminded of the necessity of good English and orderly presentation of your material.*

SECTION A: SYSTEM ARCHITECTURE (40 MARKS)

Question 1 (20 Marks)

A management information system (MIS) is an organized integration of hardware and software technologies, data, processes, and human elements designed to produce timely, integrated, relevant, accurate, and useful information for decision-making purposes. In addition to hardware, software, and human elements, an information system includes four major components

Work required

- 1) Name the four components (2 marks)
- 2) Give the definition of each component. (4 marks)
- 3) Contrast information systems and Information Technologies. (10 marks)
- 4) Name FOUR (04) examples of information technologies (4 marks)

Question 2 (10 Marks)

The new accounting system is operational, but feedback from users has been negative. The most common complaint is that the system is not user-friendly. Some people in the IT department think that more user training would solve the problem. However, Sam, the IT manager, is opposed to a fresh round of training. "Let's just set up the network to monitor the users' keystrokes and mouse clicks, and see what the patterns are," he suggested. "We can analyze the data and come up with tips and suggestions that would make the system easier to use." Your initial reaction is that Sam is wrong, for two reasons. First, you believe that monitoring would not be an effective method to learn what users really want. In your view, that should have been done in the system requirements phase. Second, you are bothered by an ethical question: Even though the proposed monitoring would involve company business, the company network, and company time, you feel that many users would resent the unannounced monitoring and might feel that their performance or other computing activities were being appraised without their knowledge. Sam has asked to you to write up a recommendation.

Work required

What will you say about the ethical question that troubles you?

(10 marks)

Question 3

(10 Marks)

- a) In the context of network, what is a node? Give three (03) examples of nodes. (2 mark)
- b) Explain the context of scalability in the context of system architecture. (2 marks)
- c) Define the following concepts:
 - i. client, server
 - ii. client/server
 - iii. system architecture
- d) What are the main components of system architecture?

(3 × 1 = 3 marks)

(3 marks)

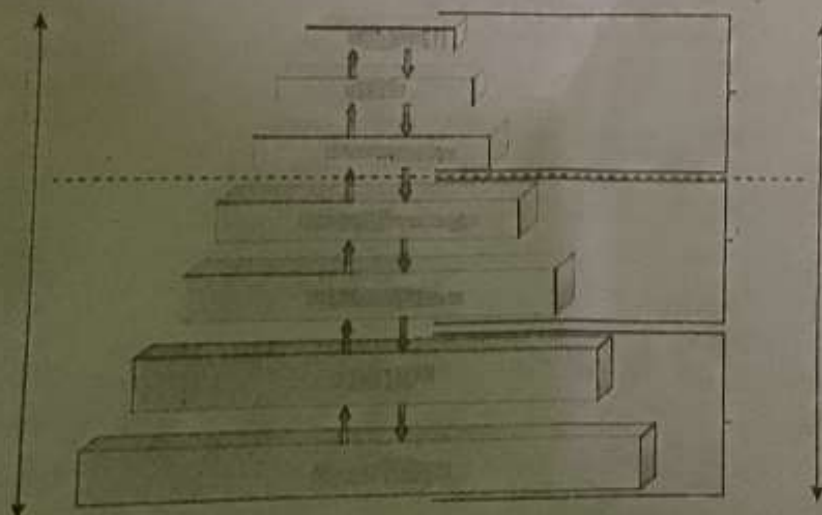
SECTION B: OPERATING SYSTEMS (40 MARKS)

Question I:

(10 Marks)

Study the figure below describing the storage device hierarchy, using the following terms to put on the figure:

Storage capacity, Smaller, Faster, volatile storage, nonvolatile storage, Registers, Cache, main memory, nonvolatile memory, hard-disk drives, optical disk, magnetic tapes, primary storage, secondary storage, tertiary storage, access time and slower



Question II:

(10 Marks)

MCQ One (01) For Each Correct Answer

1. DRAM stands for _____

- (a) digital random-access memory
- (b) dynamic random-access memory
- (c) dual random-access memory
- (d) data random access memory.

2. Computer architecture is defined by _____
(a) its word size
(b) its DRAM capacity
(c) its processor speed
(d) none of these answers.
3. An operating system is _____
(a) a low-level software that installs applications
(b) a low-level software that coordinates and schedules all tasks or processes in the computer
(c) a low-level language that coordinate and schedules computer peripherals
(d) none of these answers.
4. Three common operating systems are
(a) Microsoft Windows, Apple macOS, Python
(b) Apple iOS, Linux OS, Java Script
(c) Google's Android OS, Apple iOS, Apple macOS
(d) Windows, Apple macOS, Blackberry.
5. Two operating systems for smart phones are:
(a) Apple iOS, Windows Phone
(b) Android OS, Linux OS
(c) Python, Apple iOS
(d) Windows, Apple macOS.
6. Three issues related to micro architecture are:
(a) pipelining, cache memory, sequential processing
(b) cache memory, DRAM, CPU registers
(c) cache memory, pipelining, superscalar
(d) pipelining, sequential processing, superscalar.
7. Instructions per second (IPS) can be influenced by the following except:
(a) pipelining
(b) clock speed
(c) multi- stage pipelining
(d) sequential processing.
8. Information or data are stored in a computer installed with the Windows OS using:
(a) folders and documents
(b) spreadsheets and files
(c) files and folders
(d) folders and directories
9. Files or folders can be _____
(a) deleted
(b) renamed
(c) copied
(d) all these answers are correct.
10. An application _____ should have a dot (.) extension
(a) folder
(b) file
(c) Java
(d) Linux

Question III

(10 Marks)

Although there are many practitioners of computer science, only a small percentage of them will be involved in the creation or modification of an operating system. Why, then, study operating systems and how they work? (10 marks)

SECTION C: PROJECT MANAGEMENT

(20 MARKS)

Question IV

(20 Marks)

1. What is a project? (1 mark)
2. Define software project management (1 mark)
3. What are the characteristics of a project? (3 marks)
4. What are the characteristics that make software projects different from other projects? (3 marks)
5. What are the three successive processes that bring a new system (3 marks)
6. What are the phases in software development life cycle? (3 marks)
7. List the various ways to categorize software projects (3 marks)
8. Who are project stakeholders? (1 mark)
9. What is project steering committee? What are their roles? (2 marks)